AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2018

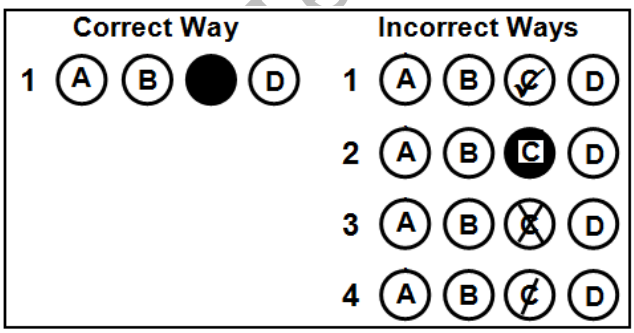
Business Statistics Paper I

Time: 30 minutes Marks: 20

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 20 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. One of the examples of quantitative data is the

   A. brand of a perfume you wear.
   B. hair colour of a group of friends.
   C. average mileage of your car per week.
   D. variety of frozen seafood available in a supermarket.

2. Which of the following options MOST suitably refers to inferential statistics?

   A. It is used to describe data.
   B. It is used to represent data.
   C. It is used to summarise data.
   D. It is used to make conclusion about data.

3. Which of the following options is representing qualitative data?

   I. Smell and taste
   II. Feel and texture
   III. Height and weight

   A. II only
   B. III only
   C. I and II
   D. II and III

4. The following table presents the height (in centimetre) of 60 students in a school.

Height (cm)	Number of Student
110 – 119	12
120 – 129	15
130 – 139	18
140 – 149	10
150 – 159	5

   In the given table, the relative frequency corresponding to the class interval 120 – 129 is

   A. 0.038
   B. 0.25
   C. 15
   D. 27

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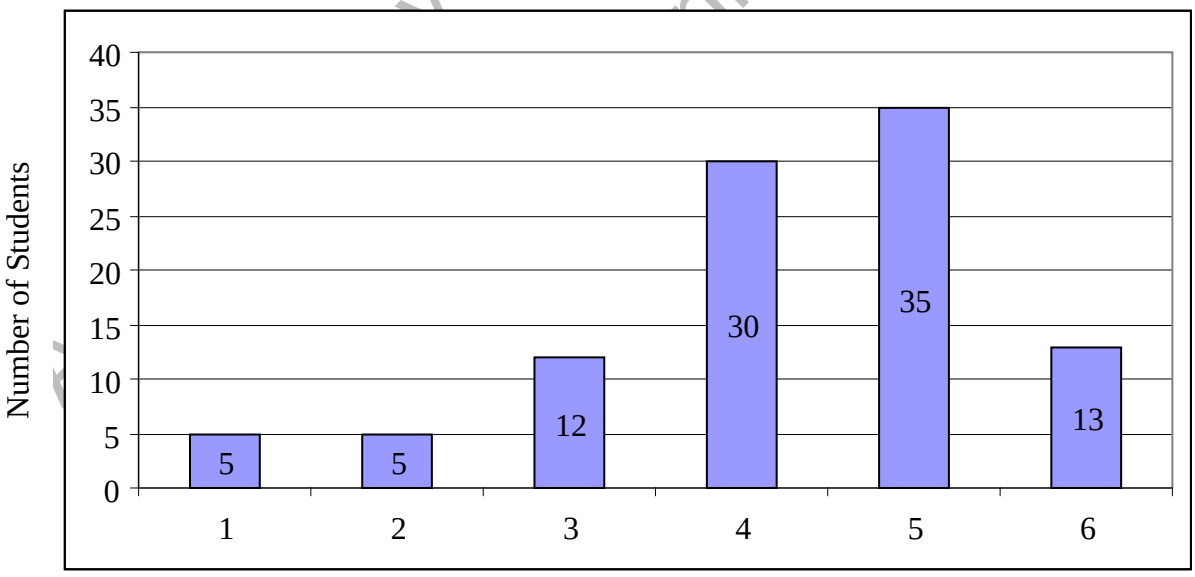
5. The following table represents the age (in years) of 30 participants in an event.

Age of Participant (Year)	Number of Participant
10 – 13	6
14 – 17	8
18 – 21	11
22 – 25	3
26 – 29	2

How many participants have more than 17 years of age?

A. 8
B. 11
C. 14
D. 16

6. The graph represents data about the number of courses opted by 100 students in an MBA class.

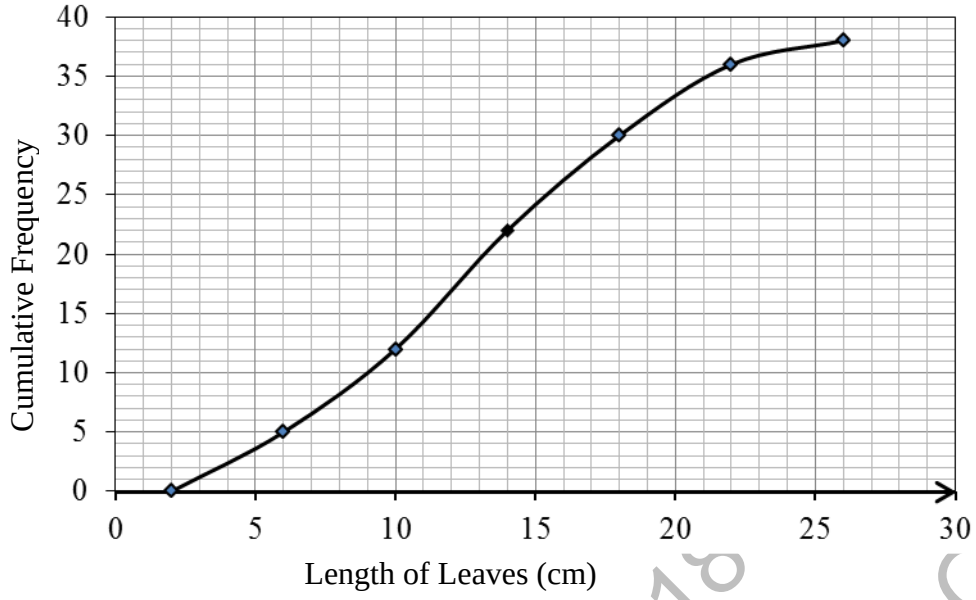


The number of students who opted for fewer than 3 courses is

A. 30
B. 22
C. 10
D. 5

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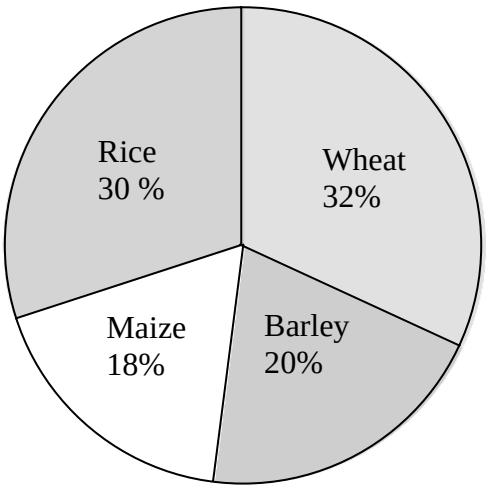
7. A botanist studied certain species of a plant and recorded length (in centimetres) of the leaves. On the basis of this data, a cumulative frequency curve is drawn.



How many leaves have length more than 14 cm?

A. 12
B. 16
C. 22
D. 38

8. A shopkeeper has four commodities, maize, rice, wheat and barley as represented in the given pie chart. Each commodity is measured in kilogram.



What is the angle associated with wheat in the given pie chart, which is 32% of the total commodities?

A. 4.44^{o}
B. 8.89^{o}
C. 57.6^{o}
D. 115.2^{o}

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9. The average driving speed of 8 vehicles on a highway is *a* km/ h. If due to rain, every vehicle slows down by *b* km/ h, then the new average driving speed will be

A. $a - b$ km/ h
B. $a + b$ km/ h
C. $\frac{a-b}{8}$ km/ h
D. $\frac{a+b}{8}$ km/ h

10. The arithmetic mean of a, b, c, d and e is 23. If $a + b + c = 90$, then $d + e$ is equal to

A. 23
B. 25
C. 44
D. 67

The organisers of a swimming competition recorded the ages of the participants shown in the given table.

Ages of Participant (years)	Number of Participant
10 – 16	16
17 – 23	18
24 – 30	56
31 – 37	13
38 – 44	7

Using the given information, answer Q.11. and Q.12.

11. The cumulative frequency preceding the median age group will be

A. 18
B. 34
C. 56
D. 90

12. The class boundaries of the modal age group is

A. 37.5 – 44.5
B. 23.5 – 30.5
C. 38.5 – 44.5
D. 24.5 – 30.5

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13. The sum of deviation of a data set about its mean is

A. zero.
B. negative.
C. mean deviation.
D. standard deviation.

14. A data set consists of ten numbers. If the sum of these numbers is 55 and sum of square of these numbers is 385, then the variance in the data set will be

A. 2.87
B. 8.25
C. 8.30
D. 68.75

15. The price index for sugar was 82.3 in year 2016, taking year 2014 as base period. This implies that the price of sugar was

A. 82.3% higher in 2016 than in 2014.
B. 82.3% higher in 2014 than in 2016.
C. 17.7% lower in 2016 than in 2014.
D. 17.7% lower in 2014 than in 2016.

16. The price of rice per kg for the years 2010 to 2014 are given in the following table along with their link relatives. The base year is 2010.

Year	Price (Rs)	Link Relative
2010	80	100
2011	85	106.25
2012	90	105.88
2013	97	107.78
2014	105	108.25

Using chain base method, the price index of rice for the year 2013 is

A. 114.12
B. 116.67
C. 118.12
D. 119.67

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17. The number of ways a coach can select a cricket team of 12 members from a group of 15 players is

A. 180
B. 455
C. 479,001,600
D. 217,945,728,000

18. How many ways letters of the word 'TRAIN' can be arranged if all the letters are used once?

A. 25
B. 32
C. 120
D. 125

A box contains 5 red, 7 black and 8 white balls. Two balls are drawn at random.

Using the given information, answer Q.19 and Q.20.

19. If the balls are drawn with replacement, then the probability that the first ball is red and the second ball is white will be

A. $\frac{5}{20} \times \frac{8}{20}$
B. $\frac{5}{20} + \frac{8}{20}$
C. $\frac{5}{20} \times \frac{8}{19}$
D. $\frac{5}{20} + \frac{8}{19}$

20. If the balls are drawn without replacement, then the probability that both the balls are black will be

A. $\frac{1}{7} \times \frac{1}{7}$
B. $\frac{1}{7} \times \frac{1}{6}$
C. $\frac{7}{20} \times \frac{6}{19}$
D. $\frac{7}{20} \times \frac{7}{20}$

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END OF PAPER

Please use this page for rough work

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